

standard LP form using the above defined matrices. We will now solve the above problem using Octave. The Octave function *glpk* is used to solve LP problems. Octave uses the GNU GLPK (www.gnu.org/software/glpk) library, hence the function name.

The syntax of the function is: *glpk(c, a, b, lb, ub, ctype, vartype, sense, param)*. For this particular problem, we just have the *c, a, b* parameters. So, here we go in Octave:

```
octave:1> c = [-2,-1]
C =

-2  -1

octave:2> A = [2, -1; 1, 2]
A =

2  -1
1  2

octave:3> b = [8; 14]
B =

8
14

octave:4> glpk ( c, A, b )
ans =

6
4
```

So, we have the solution to the given LP. Simple! (Note that we have defined the transpose of *c* as *c* since *glpk* expects the transpose as shown in the standard LP form.)

Among the other parameters, *sense* tells *glpk* whether the

problem is a minimisation or a maximisation problem. A value of 1 indicates a minimisation problem and a value of -1 indicates a maximisation problem. The default value is 1. Usage of other parameters will vary from problem to problem.

Matrix factorisation

Matrix factorisation or decomposition techniques decompose a matrix into a product of simpler matrices. These techniques are very useful for implementing efficient and fast matrix algorithms. Octave has support for the LU, SV D, QR and schur decompositions, among others. Going into the details is outside the scope of this article. For the Octave functions, please refer to the Octave user manual.

Looking ahead

I hope I have been able to give you some guidance in using Octave for your linear algebraic needs in the first two articles in this series. Starting from the next instalment, I will be exploring other areas of Octave. Please feel free to contact me with your queries and suggestions! 

Resources

- GNU Octave: www.gnu.org/software/octave
- GNU Octave Function reference: <http://octave.sourceforge.net/doc/index.html>
- Wikipedia articles on matrix decomposition, linear programming and optimisation in mathematics

Acknowledgements

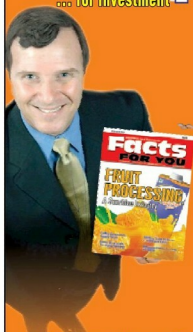
Considering the heavy usage of math formulas, LaTeX was an automatic choice to write this article and thanks to the The Not So Short Introduction to LaTeX (<http://ctan.tug.org/tex-archive/info/short/english/short.pdf>) for its nice treatment on how to write math well in LaTeX. Hevea (<http://paullac.inria.fr/hevea/>) was used for LaTeX-to-HTML conversion and it is awesome.

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